Quiz

Quiz ?

Statement

Which of the following scenarios is better suited for a SQL database?

Select Option

○ A social media application with diverse and unpredictable user-generated content.

Correct

● A banking application requiring transactional consistency.

○ A large-scale IoT application requiring high write throughput.

○ A real-time analytics application with high volume and velocity of data.

Check Answer

Quiz ?

Statement

Which type of database is more suitable for a scenario requiring horizontal scalability?

Select Option

○ SQL

Correct

● NoSQL

○ Both SQL and NoSQL

○ Neither SQL nor NoSQL

Check Answer

Quiz ?

Statement

For an application that requires complex query and aggregation operations, which type of database is more appropriate?

Select Option

Correct

● SQL

○ NoSQL

○ Both SQL and NoSQL

○ Neither SQL nor NoSQL

Check Answer

Quiz ?

Statement

What is the primary purpose of the Domain Name System (DNS)?

Select Option

○ Encrypting web traffic

○ Compressing web content

Correct

● Translating domain names to IP addresses

Explanation

The primary purpose of DNS is to translate human-readable domain names into machine-readable IP addresses, allowing users to access websites and online services using memorable domain names instead of having to memorize numerical IP addresses.

○ Routing email messages

Check Answer

Quiz ?

Statement

Which DNS server holds the actual DNS records for a domain?

Select Option

○ Root server

○ TLD server

Correct

● Authoritative name server

Explanation

Authoritative name servers hold the actual DNS records for a domain, including its IP address and other information. They provide the final answer to DNS queries, enabling users to access the desired website or resource.

○ DNS resolver

Check Answer

Quiz ?

Statement

What is the Time To Live (TTL) in a DNS record?

Select Option

Correct

● The time a DNS record is stored in the cache before being removed

Explanation

Time To Live (TTL) is a value associated with a DNS record, specifying how long the record should be stored in the cache. TTL is measured in seconds, and once the TTL expires, the cached information is removed to ensure that outdated information is not used.

○ The time it takes for a DNS query to reach the server

○ The time it takes for a DNS server to process a query

○ The time it takes for a DNS query to travel across the internet

Check Answer

Quiz ?

Statement

Which technique allows multiple servers to share the same IP address and routes users to the nearest server based on network latency and server availability?

Select Option

○ Round-robin DNS

○ Geographically distributed DNS servers

○ Content Delivery Networks (CDNs)

Correct

● Anycast routing

Explanation

Anycast routing is a networking technique that allows multiple servers to share the same IP address. It routes users' queries to the nearest server based on factors like network latency and server availability, providing load balancing, reduced latency, and high availability for DNS services.

Check Answer

Quiz ?

Statement

What is the primary purpose of caching in computer systems?

Select Option

○ To encrypt data

Correct

● To store data temporarily for faster access

Explanation

The primary purpose of caching is to store data temporarily for faster access. By caching frequently accessed data, systems can reduce the need to repeatedly fetch the same information from the original source, thereby improving performance and reducing resource usage.

○ To permanently store data

○ To compress data

Check Answer

Quiz ?

Statement

Which of the following is NOT a common type of cache?

Select Option

○ CPU cache

○ Browser cache

○ Disk cache

Correct

● Time cache

Explanation

Time cache is not a common type of cache. Common types of cache include CPU cache (used by processors to store frequently accessed data), browser cache (used by web browsers to store web content), and disk cache (used by storage devices to temporarily store data for faster access).

Check Answer

Quiz ?

Statement

What is the primary downside of caching?

Select Option

○ Increased security risk

○ Increased storage space requirements

Correct

● The potential for serving outdated or stale data

Explanation

The primary downside of caching is the potential for serving outdated or stale data. Since cached data is a copy of the original data, any changes made to the original data may not be reflected in the cache immediately. This can result in users accessing outdated information until the cache is updated or the cached data expires.

○ Slower access to data

Check Answer

Quiz ?

Statement

Which cache replacement policy removes the item that has not been accessed for the longest time?

Select Option

Correct

● Least Recently Used (LRU)

Explanation

The Least Recently Used (LRU) cache replacement policy removes the item that has not been accessed for the longest time. This policy aims to keep frequently accessed data in the cache while evicting less frequently used data to make room for new items.

○ First In, First Out (FIFO)

○ Most Recently Used (MRU)

○ Random Replacement (RR)

Check Answer

Quiz ?

Statement

What is a Content Delivery Network (CDN) cache primarily used for?

Select Option

○ Storing data on the user's local device

Correct

● Reducing the load on the main server by distributing content across multiple servers

Explanation

A Content Delivery Network (CDN) cache is primarily used to reduce the load on the main server by distributing content across multiple servers. CDNs store copies of web content on geographically distributed servers, allowing users to access the content from a server that is closer to them, resulting in faster load times and improved performance.

○ Storing frequently accessed files on a single server

○ Encrypting data before it is sent across the internet

Check Answer

Quiz ?

Statement

In caching, what does "cache hit" mean?

Select Option

○ The process of loading data into the cache

○ An error that occurs when the cache is full

Correct

● When the requested data is found in the cache

Explanation

A "cache hit" occurs when the requested data is found in the cache. This means that the system can quickly access the data without having to fetch it from the original source, resulting in faster performance and reduced resource usage.

○ When the requested data is not found in the cache

Check Answer

Quiz ?

Statement

Which of the following is NOT a benefit provided by proxy servers?

Select Option

○ Accelerating web page loading times through caching

○ Improving network security by filtering malicious content

○ Bypassing network restrictions or geolocation-based content blocking

Correct

● Increasing storage capacity on client devices

Explanation

Proxy servers do not increase storage capacity on client devices. Their main benefits include caching, improving network security, load balancing, and bypassing network restrictions or geolocation-based content blocking.

Check Answer

Quiz ?

Statement

Which type of proxy is commonly used to provide anonymity and protect users' privacy?

Select Option

○ Transparent proxy

○ Reverse proxy

Correct

● Forward proxy

Explanation

Forward proxies, also known as regular proxies or just proxies, are commonly used to provide anonymity and protect users' privacy. By routing client requests through a forward proxy, the destination server sees the proxy's IP address instead of the user's, helping to mask the user's identity and maintain privacy.

○ Caching proxy

Check Answer

Quiz ?

Statement

In horizontal partitioning, how is data typically divided among nodes?

Select Option

Correct

● By rows

Explanation

In horizontal partitioning, also known as sharding, data is typically divided among nodes by rows. Each node is responsible for a specific range or subset of the data, which can help distribute load and improve performance.

○ By columns

○ By tables

○ By databases

Check Answer

Quiz ?

Statement

What is the main advantage of vertical partitioning?

Select Option

○ It reduces the size of the data stored on each node

Correct

● It improves query performance by storing related columns together

Explanation

The main advantage of vertical partitioning is that it improves query performance by storing related columns together on the same node. This approach can help reduce the amount of data that needs to be transferred between nodes when executing queries, leading to faster query execution times.

○ It distributes load evenly across all nodes in the system

○ It encrypts data before storing it

Check Answer

Quiz ?

Statement

Which of the following is a characteristic of Bloom filters?

Select Option

○ They provide a definitive answer for set membership

Correct

● They can produce false positives

Explanation

Bloom filters can produce false positives, which means they might indicate that an element is a member of a set when it is not. However, they do not produce false negatives, meaning if the filter indicates an element is not in the set, it is guaranteed not to be.

○ They produce false negatives

○ They require a large amount of memory

Check Answer

Quiz ?

Statement

In a Bloom filter, what is the purpose of using multiple hash functions?

Select Option

○ To compress data

Correct

● To reduce the likelihood of false positives

Explanation

In a Bloom filter, multiple hash functions are used to reduce the likelihood of false positives. By using different hash functions, the filter can distribute the set membership information more evenly across the bit array, making it less likely that two different elements will produce the same set of hash values.

○ To increase the likelihood of false positives

○ To encrypt data

Check Answer

Quiz ?

Statement

Which of the following is an advantage of using a distributed messaging queue in a system?

Select Option

○ Reduced latency for all operations

○ Guaranteed message delivery in all cases

Correct

● Improved fault tolerance and system resilience

Explanation

Using a distributed messaging queue in a system can improve fault tolerance and system resilience. It allows components to send and receive messages without being directly connected, which means that if one component fails, the others can still continue to operate. Additionally, messaging queues can often buffer messages, ensuring that they are not lost if a component is temporarily unavailable.

○ Total elimination of data loss

Check Answer

← Previous

Event-Driven Vs. Polling Architecture

☐ Mark as Completed